The Carbon Cycle – extracting information from a text

Instructions

- Read the short text below;
- Complete the tasks underneath.

"Living things are mostly made up from a handful of elements. Carbon is one of the most important of these. Carbon is found in the bodies of animals and in plants.

Plants get the carbon from the atmosphere in the form of carbon dioxide. They take this in through their leaves. It is formed into sugars in the plant by the process of photosynthesis. Some of the sugars are used to provide the plant with energy in the process of respiration, which releases carbon dioxide into the atmosphere. They convert some of the sugars to other carbohydrates, which form the structure of the plant. When the plants die, the structural carbohydrates are broken down by decomposers in the process of decomposition. This releases the carbon in the form of carbon dioxide into the atmosphere.

Some plants don't decompose when they die. They get trapped in underground or underwater without the oxygen that decomposers need to break them down. This has caused some of the plants that did this hundreds of millions of years ago to form fossil fuels like coal.

Fossil fuels are being dug up and are burnt in power stations as a fuel. This releases carbon dioxide back into the atmosphere in a process called combustion. The carbon may have been trapped underground for millions of years. This started on a large scale during the industrial revolution and continues to increase year after year.

Some plants get eaten by animals. The carbon from the plant in the form of sugars and starches are mostly broken down inside the body of the animal. When the animal carries out respiration, some of the carbon is released back into the atmosphere as carbon dioxide. Some of the carbon is trapped in the body of the animal as part of its tissues. Some of the carbon passes straight through the animals because it isn't digested. This is part of the animal's dung. This is broken down by decomposers.

When the animal dies, the carbon that is trapped in the body of the animal is released into the atmosphere as carbon dioxide when decomposers break the body of the animal down.

Some of the carbon dioxide from the atmosphere can dissolve in water. Phytoplankton and other plants in the ocean can carry out photosynthesis, which traps the carbon inside them. These plants also respire, like plants on land. Some of these plants die and some might end up being eaten by animals. Some of the animals make shells from carbon and other elements. When the animal dies, the shell often takes a long time to break down and sometimes this get trapped, squashed and formed into carbonaceous rocks.

Carbon dioxide is released from the water into the atmosphere. This happens more if water is warmer because the bubbles of gas don't dissolve as well in warm liquids.

Volcanic activity also releases carbon dioxide into the atmosphere from rocks deep underground."

Tasks

1. In the text above highlight in one colour (e.g. yellow) all of the places that carbon is found
2. Highlight in a different colour (e.g. green) all of the processes that take carbon from the atmosphere
3. Highlight in a third colour (e.g. pink) all of the processes that return carbon to the atmosphere
4. Now turn the text into a diagram, with boxes showing the places you find carbon. Link these boxes with labelled arrows showing processes that moves carbon from one box to another. Use the same colour code for arrows.

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